

Oxalato- and Malonato-Complexes of Trivalent Metals. II

Stabilities of Tris-oxalato Complexes of Iron, Chromium, Aluminium, and Gallium

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With 2 Figures

Summary

The stabilities of tris-oxalato complexes of trivalent metals like Fe, Cr, Al, Ga have been studied. These compounds have been found to decompose in steps by interaction with acids. The stepwise instability constants and also the overall instability constants of the reactions $\text{MOx}_3''' \rightarrow \text{MOx}_2' + \text{Ox}''$, $\text{MOx}_2' \rightarrow \text{MOx}^+ + \text{Ox}''$, $\text{MOx}^+ \rightarrow \text{M}^{3+} + \text{Ox}''$ and $\text{MOx}_3''' \rightarrow \text{M}^{3+} + 3 \text{Ox}''$ have been measured.

Various complex oxalates of trivalent metals like chromium, iron, and cobalt have been prepared and successfully resolved since the days of WERNER. WAHL¹⁾ prepared and resolved the corresponding aluminium compound $\text{Al}(\text{C}_2\text{O}_4)_3'''$, complex oxalates of gallium have also been prepared and resolved by NEOGI and DUTT²⁾. The preparation and resolution of these complexes can only speak for their octahedral configurations. However, to obtain a greater insight into their character, mode of formation and the nature of the chemical bonds involved, a quantitative study of their dissociation in solution or in other words their stabilities in solution were found necessary.

The problem of complex formation has been treated on a statistical basis by J. BJERRUM³⁾ involving all the possible intermediate stages in a reversible manner. This method has been successfully employed to the case of inner-metallic complexes by CALVIN and WILSON⁴⁾, MELLOR et al.⁵⁾, ACKERMAN et al.⁶⁾, P. RAY et al.⁷⁾ and others.

¹⁾ W. WAHL, Ber. dtsch. chem. Ges. **60**, 399 (1927).

²⁾ P. NEOGI and N. K. DUTT, J. Indian chem. Soc. **15**, 83 (1938).

³⁾ J. BJERRUM, Metalammine formation in aqueous solution. P. Haase and Sons, Copenhagen, 1941.

⁴⁾ M. CALVIN and K. W. WILSON, J. Amer. chem. Soc. **67**, 2003 (1945).

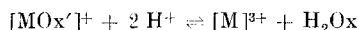
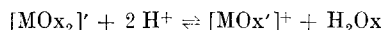
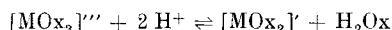
⁵⁾ D. P. MELLOR et al., J. Amer. chem. Soc. **70**, 401 (1948).

⁶⁾ H. ACKERMAN et al., Nature [London] **163**, 723 (1949).

⁷⁾ P. RAY et al., J. Indian chem. Soc. **27**, 493 (1950); *ibid.* **33**, 841 (1956).

This method has been used by us in the study of the trisoxalato complexes of trivalent metals, which forms the subject matter of the present investigation.

The tris-oxalato complexes of tripositive metals react with acids in aqueous solution in the following way



where M is trivalent metal ion like Fe, Cr, Al and Ga, and Ox'' and H₂Ox represent respectively oxalato ion and undissociated oxalic acid molecule.

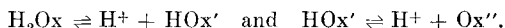
The equilibrium constants of the reactions are given by

$$k_3' \rightleftharpoons \frac{[\text{MOx}_2]' \cdot [\text{H}_2\text{Ox}]}{[\text{MOx}_3]''' \cdot [\text{H}^+]^2}; \quad k_2' \rightleftharpoons \frac{[\text{MOx}]^+ \cdot [\text{H}_2\text{Ox}]}{[\text{MOx}_2]' \cdot [\text{H}^+]^2}; \quad k_1' \rightleftharpoons \frac{[\text{M}]^{3+} \cdot [\text{H}_2\text{Ox}]}{[\text{MOx}]^+ \cdot [\text{H}^+]^2}.$$

These are related to the successive instability or dissociation constants k_3 , k_2 , k_1 , in the following way

$$k_3 = \frac{[\text{MOx}_2]' \cdot [\text{Ox}']}{[\text{MOx}_3]'''} = \frac{[\text{MOx}_2]' [\text{H}_2\text{Ox}]}{[\text{MOx}_3]''' [\text{H}^+]^2} \cdot \frac{[\text{HOx}'] [\text{H}^+]}{[\text{H}_2\text{Ox}]} \cdot \frac{[\text{Ox}'] [\text{H}^+]}{[\text{HOx}]} = k_3' \cdot k_1^* \cdot k_2^*.$$

Similarly, $k_2 = k_2' \cdot k_1^* \cdot k_2^*$ and $k_1 = k_1' \cdot k_1^* \cdot k_2^*$ where k_1^* and k_2^* are the first and second respectively acid dissociation constant of oxalic acid (K₂Ox) from



The overall dissociation constant of the reaction



is given by $K = k_3 \cdot k_2 \cdot k_1 = k_3' \cdot k_2' \cdot k_1' \cdot k_1^{*3} \cdot k_2^{*3}$.

The acid dissociation constants (k_1^* and k_2^*) of oxalic acid were determined from a measurement of the p_{H} values of the equilibrium mixture of a known amount of sodium oxalate and definite volumes of standard acid (HNO₃) of widely varying strengths at a constant temperature. Solutions were made in molar KNO₃.

Oxalic acid in acid solution is present as Ox'', HOx' and H₂Ox.

$$k_1^* = \frac{A_1 \cdot [\text{H}^+]}{A_2} = \frac{(2A - C_A + C'_A) [\text{H}^+]}{(C_A - A - C'_A)}$$

$$k_2^* = \frac{A_0 \cdot [\text{H}^+]}{A_1} = \frac{(A - C_A + C'_A) [\text{H}^+]}{(C_A - C'_A)},$$

where A = total oxalate taken,

A_2 = conc. of undissociated oxalic acid in solution,

A_1 = conc. of HOx' in solution,

A_0 = conc. of Ox'' in solution

C_A = conc. of the acid added,

C'_A = conc. of the acid after reaction, from the final p_H values of the solution mixture and the p_H - acid relation graph (RANDALL and YOUNG⁸).

$A = A_0 + A_1 + A_2$

$C_A - C'_A$ = acid consumed
 $= A_1 + 2 A_2$.

Evaluations of k'_3 , k'_2 , k'_1 were similarly made from the measurements of the p_H values of the equilibrium mixture of a known amount of the complex and definite volumes of standard nitric acid of varying concentrations at a constant temperature. Here also, as before, solutions were made with molar KNO_3 .

The calculation is made as follows:

Let C denotes the molar concentration of the complex taken,

C_3 denotes the molar concentration of $[\text{MOx}_3]'''$ at equilibrium

C_2 denotes the molar concentration of $[\text{MOx}_2]'$ at equilibrium

C_1 denotes the molar concentration of $[\text{MOx}]^+$ at equilibrium

C_0 denotes the molar concentration of $[\text{M}^{3+}]$ at equilibrium

X denotes the oxalic acid liberated from the complex $3C - X = 3C_3 + 2C_2 + C_1$ = oxalate attached to metal, Z = acid consumed in the process at equilibrium by the reaction of the acid with the complex, obtained from the concentration of the acid added and the final p_H of the solution mixture and the p_H -acid relation graph.

= acid added - acid left.

= $2 A_2 + A_1$

= $A_2 \cdot \frac{2 [\text{H}^+] + k_1^*}{[\text{H}^+]}$

$C = C_3 + C_2 + C_1 + C_0$

whence, $X = \frac{Z ([\text{H}^+]^2 + k_1^* [\text{H}^+] + k_1^* \cdot k_2^*)}{[\text{H}^+] (2 [\text{H}^+] + k_1^*)}$.

Three cases are possible depending upon the relative values of X and C

1. $X < C$, C_1 and C_0 are negligibly small or $C = C_3 + C_2$ when,

$$k'_3 = \frac{X}{C - X} \cdot \frac{Z}{[\text{H}^+] (2 [\text{H}^+] + k_1^*)} = \frac{X}{C - X} \cdot \frac{A_2}{[\text{H}^+]^2}.$$

2. $X > C > 2C$, C_3 and C_0 are negligibly small or $C = C_2 + C_1$, when,

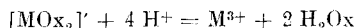
$$k'_2 = \frac{X - C}{2C - X} \cdot \frac{A_2}{[\text{H}^+]^2}.$$

3. $X > 2C < 3C$, C_3 and C_2 are negligibly small or $C = C_1 + C_0$

$$k'_1 = \frac{X - 2C}{3C - X} \cdot \frac{A_2}{[\text{H}^+]^2}.$$

⁸) M. RANDALL and L. E. YOUNG, J. Amer. chem. Soc. **50**, 989 (1928).

4. A fourth case shall also be considered. In this case, it is assumed that mono-oxalato ion i. e. $[\text{MOx}]^+$ does not exist in solution but the bis-oxalato ion directly breaks up into M^{3+} and H_2Ox according as



when $X > C_1$, C_3 is negligible, $C = C_0 + C_2$

$$k'' = \frac{X - C}{3C - X} \cdot \left[\frac{A_2}{[\text{H}^+]^2} \right]^2.$$

Thus the equilibrium constants and hence the instability constants are calculated from experimental data.

The experimental values were then compared with the graphical values derived according to the method of CARLSON, McREYNOLD and VERHOEK⁹).

$\bar{n} = \frac{3C - X}{C}$ where $\bar{n}$ is the ratio of oxalic acid still bound to the metal to the concentration of the complex taken originally, called formation function by BJERRUM,

$$\begin{aligned} &= \frac{[\text{MOx}]^+ + 2[\text{MOx}_2]' + 3[\text{MOx}_3]'''}{[\text{M}^{3+}] + [\text{MOx}]^+ + [\text{MOx}_2]' + [\text{MOx}_3]'''} \\ &= \frac{Y_1[b] + 2Y_1Y_2[b]^2 + 3Y_1Y_2Y_3[b]^3}{1 + Y_1[b] + Y_1Y_2[b]^2 + Y_1Y_2Y_3[b]^3} \end{aligned} \quad (\text{A})$$

when $Y_1 = \frac{1}{k'_1}$, $Y_2 = \frac{1}{k'_2}$, $Y_3 = \frac{1}{k'_3}$ and $[b] = \frac{[\text{H}_2\text{Ox}]}{[\text{H}^+]^2} = \frac{A_2}{[\text{H}^+]^2}$.

Again, when $n = n - \frac{1}{2}$, there will be about equal amounts of MOx_{n-1} , and MOx_n in the equilibrium reaction.

So that as a first approximation,

$$\frac{1}{k'_n} = Y_{\bar{n}} = \frac{1}{[b]_{\bar{n} = n - \frac{1}{2}}}.$$

Now, solving eq. (A) for individual constants and substituting this value of Y_n in the solutions, we get more accurately,

$$\begin{aligned} Y_1 &= \frac{1}{[b]_{\bar{n} = \frac{1}{2}}} = \frac{1}{1 + 3Y_2[b]_{\bar{n} = \frac{1}{2}}} = \frac{1}{\frac{1}{2} + 5Y_2Y_3[b]^2_{\bar{n} = \frac{1}{2}}} \\ Y_2 &= \frac{1}{[b]_{\bar{n} = \frac{3}{2}}} = \frac{1}{1 + 3Y_3[b]_{\bar{n} = \frac{3}{2}}} = \frac{1}{\frac{3}{2} + \frac{3}{Y_1[b]_{\bar{n} = \frac{3}{2}}}} \\ Y_3 &= \frac{1}{[b]_{\bar{n} = \frac{5}{2}}} = \frac{1}{\frac{5}{2} + \frac{3}{Y_2[b]_{\bar{n} = \frac{5}{2}}} + \frac{5}{Y_1Y_2[b]^2_{\bar{n} = \frac{5}{2}}}} \end{aligned}$$

The values of $[b]$ are obtained from the curve drawn by plotting $\bar{n}$ against $\log [b]$. Thus Y_1, Y_2, Y_3 may be obtained approximately from the values of $[b]$ when $\bar{n} = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}$ respectively and hence k'_1, k'_2, k'_3 are computed.

⁹) G. A. CARLSON, J. P. McREYNOLDS, F. H. VERHOEK, J. Amer. chem. Soc. **67**, 1334 (1945).

All solutions employed were prepared with conductivity water. A BECKMANN p_H -meter, model G was used for p_H measurements. Solutions were kept in an electrically regulated thermostat for several hours before measurements till equilibrium was attained. Complex salts used were prepared by known methods.

Discussion of results

The first and second dissociation constants of oxalic acid were found to be $7.586 \cdot 10^{-2}$ (k_1^*) and $2.089 \cdot 10^{-4}$ (k_2^*), respectively at 32°C . A perusal of the results tabulated in tables II–V and shown graphically

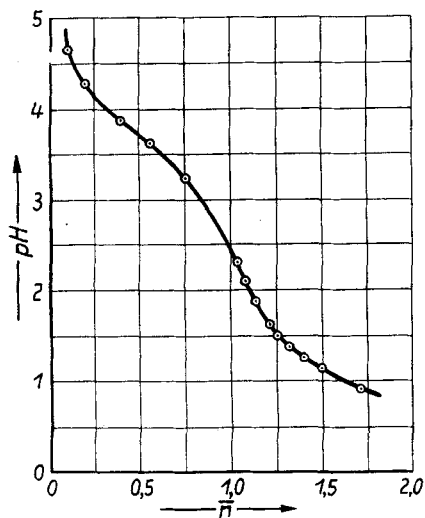


Fig. 1

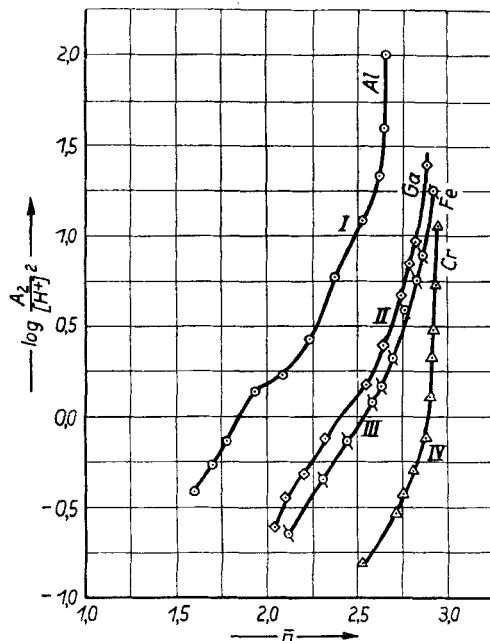
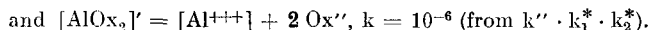
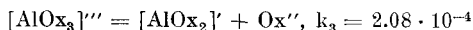


Fig. 2

in figure 2 shows that oxalic acid forms complexes of moderate stabilities with trivalent metals like aluminium, gallium, iron and chromium. For aluminium, we could detect two equilibria



The decomposition of bis-oxalato complex to mono-oxalato complex cation i. e. $[\text{AlOx}]^+$ has also been inferred from the experiments as will be evident from the results in table column heading k_2' which also give good concordant results as good as those under the head k'' . LACROIX¹⁰⁾ failed to detect the presence of any $[\text{AlOx}]^+$ by his method of

¹⁰⁾ S. LACROIX, Ann. Chim. 4, 1 (1949).

Table I

Sodium Oxalate = 0.10 M in M KNO₃; C_A = acid added, C'_A = acid left obtained from p_H (from the p_H-C_H graph), t = 32° C

C _A	p _H	C' _A · 10 ⁴	P k ₁ [*] · 10	P k ₂ [*]	$\bar{n}$
0.001	4.645	0.23		3.68	0.10
0.002	4.290	0.51		3.67	0.20
0.004	3.870	1.43		3.67	0.39
0.006	3.620	2.51		3.65	0.58
0.008	3.230	6.31		3.65	0.74
0.012	2.640	25.1		3.91	0.95
0.016	2.295	56.2	8.92		1.04
0.02	2.080	93.3	9.36		1.07
0.03	1.780	186.2	9.84		1.14
0.04	1.595	278.6	10.30		1.21
0.05	1.480	371.5	10.80		1.28
0.06	1.385	467.7	10.64		1.32
0.08	1.240	661.0	10.46		1.39
0.10	1.130	851.0	11.13		1.49
0.16	0.905	1429	12.35		1.71

The following values for k₂^{*} and k₁^{*} were taken for calculations, as found from curve 2. k₂^{*} = 2.089 · 10⁻⁴; k₁^{*} = 7.586 · 10⁻²; k₁^{*} k₂^{*} = 1.585 · 10⁻⁵.

Table II

Na₃[AlOx₃] 3 H₂O Conc. = 0.01 M in M KNO₃; C_A = acid added, C'_A-acid left obtained from p_H-C_H curve X-total oxalate liberated from the complex, t = 32° C

C _A	p _H	C' _A · 10 ³	log $\frac{A_2}{[H^+]^2}$	X · 10 ³	k ₂ ' · 10	k ₃ '	$\bar{n}$	K''
0.002	3.64	0.2	2.003	3.35		50.6	2.67	
0.004	3.00	1.05	1.59	3.56		21.7	2.64	
0.006	2.66	2.35	1.318	3.86		13.0	2.62	
0.010	2.33	5.13	1.088	4.80		11.3	2.52	
0.020	1.925	13.0	0.774	6.23		9.8	2.38	
0.040	1.560	32.2	0.434	7.64		8.8	2.24	
0.060	1.365	47.9	0.240	8.95		14.7	2.11	
0.080	1.240	64.6	0.148	10.56	0.66		1.94	0.057
0.1205	1.050	101.0	-0.134	12.15	2.01		1.78	0.065
0.1607	0.915	140.0	-0.266	13.00	2.32		1.70	0.052
0.201	0.810	178.0	-0.410	13.91	2.43		1.61	0.087

k₃' (mean) = 12.55 (omitting borderline doubtful value 50.6)

k₂' (mean) = 2.25 · 10⁻¹ (omitting the value 0.66 · 10⁻¹).

k₂ (instability const.) = K₂' · k₁^{*} · k₂^{*} = 3.57 · 10⁻⁶, k₃ = 2.08 · 10⁻⁴.

K'' = 5.8 · 10⁻².

From curve I figure 2,

Y₃ = 0.0871; k₃' = $\frac{1}{0.0871}$ = 11.48, after correction, the value becomes 11.85.

Y₂ = 4.365; k₂' = $\frac{1}{4.365}$ = 2.29 · 10⁻¹, after correction, it becomes 2.32 · 10⁻¹.

Table III
 $K_3[Ga(Ox)_3] \cdot 3 H_2O$ Temp. = $32^\circ C$
 Conc. —0.01 M in $M KNO_3$

C_A	P_H	$C'_A \cdot 10^3$	$\log \frac{A_2}{[H^+]^2}$	$X \cdot 10^3$	k'_3	$\bar{n}$
0.001	3.47	0.365	1.3967	1.04	2.602	2.89
0.002	3.055	0.933	1.934	1.30	2.336	2.87
0.004	2.66	2.34	0.9750	1.76	2.024	2.82
0.006	2.44	3.89	0.8447	2.13	1.89	2.79
0.01	2.19	7.08	0.6875	2.66	1.769	2.73
0.02	1.835	16.0	0.4014	3.37	1.282	2.66
0.04	1.51	33.9	0.1584	4.77	1.310	2.52
0.06	1.33	71.9	0.128	5.91	1.475	2.41
0.08	1.205	70.0	—0.1079	6.77	1.559	2.33
0.1205	1.025	108.0	—0.3141	7.79	1.707	2.22
0.1607	0.895	145	—0.4621	8.94	2.89	2.11
0.2009	0.795	184	—0.5998	9.51	4.86	2.05

k'_3 (mean) = 1.795 (omitting borderline values 2.602 and 4.86),

k_3 (instability constant) = $k'_3 \cdot k_1^* \cdot k_2^* = 2.85 \cdot 10^{-5}$.

From curve II figure 2,

$$Y_3 = 0.72; k'_3 = \frac{1}{0.72} = 1.39.$$

Table IV
 $K_3[Fe(Ox)_3] \cdot 3 H_2O$ Temp. = $32^\circ C$
 Conc. —0.01 M in $M KNO_3$

C_A	P_H	$C'_A \cdot 10^3$	$\log \frac{A_2}{[H^+]^2}$	$X \cdot 10^3$	k'_3	$\bar{n}$
0.001	3.38	0.44	1.2465	0.84	1.615	2.92
0.002	3.00	1.07	1.0762	1.11	1.48	2.89
0.004	2.63	2.51	0.8993	1.54	1.442	2.85
0.006	2.42	4.17	0.7613	1.84	1.301	2.82
0.01	2.16	7.59	0.5901	2.23	1.155	2.78
0.02	1.32	16.6	0.3265	2.95	0.889	2.70
0.03	1.63	25.7	0.1750	3.50	0.807	2.65
0.04	1.50	34.7	0.0833	4.13	0.8535	2.59
0.06	1.31	33.7	—0.1297	5.33	0.8476	2.47
0.08	1.20	70.8	—0.1405	6.34	1.251	2.37
0.12	1.01	110	—0.3367	6.91	1.058	2.31
0.16	0.89	148	—0.5503	7.41	0.8041	2.26
0.201	0.79	186	—0.6338	8.83	1.754	2.12

k'_3 (mean) = 1.081 (omitting the borderline values 1.615 and 1.754).

$k_3 = k'_3 \cdot k_1^* \cdot k_2^* = 1.71 \cdot 10^{-5}$

From curve III, figure 2,

$$Y_3 = 1.047; k'_3 = 1/1.047 = 0.955.$$

Table V
 $\text{Na}_3[\text{Cr}(\text{Ox})_3] \cdot 3 \text{H}_2\text{O}$ Temp. = 32°C
 Conc. = 0.01 M in M KNO_3

C_A	P_H	$C'_A \cdot 10^3$	$\log \frac{A_2}{[\text{H}^+]^2}$	$X \cdot 10^3$	k'_3	n
0.001	3.28	0.55	1.0449	0.62	0.7360	2.948
0.002	2.87	1.44	0.7210	0.63	0.3538	2.937
0.004	2.53	3.24	0.5011	0.78	0.2713	2.921
0.006	2.33	5.13	0.3385	0.87	0.2071	2.913
0.01	2.085	9.02	0.1173	0.92	0.1325	2.908
0.02	1.775	18.6	-0.1203	1.18	0.1041	2.882
0.04	1.475	37.6	-0.2955	1.861	0.1157	2.814
0.06	1.295	54.9	-0.4601	2.228	0.1002	2.777
0.08	1.17	75.9	-0.5369	2.820	0.1157	2.718
0.12	0.995	115.0	-0.7328	3.308	0.0914	2.669
0.16	0.875	153.0	-0.8182	4.638	0.1109	2.536

k'_3 (mean) = 0.136 (omitting first two values)

$k_3 = k'_3 \cdot k_1^* \cdot k_2^* = 3.36 \cdot 10^{-6}$.

From curve IV, figure 2,

$Y_3 = 7.244$; $k'_3 = 0.138$.

formation. In case of other three metals, however, decompositions by acid do not go beyond the bis-oxalato stage, from which it appears that the bis-oxalato complexes of these metals are very stable in presence of even much acid. LACROIX l. c.) though inferred the presence of three complexes of gallium in solution, he failed to determine the dissociation constants of either of the three equilibria. We measured the dissociation constant of the first equilibrium — the only one we could detect, k_3 value is $2.85 \cdot 10^{-5}$. LAMBLING¹¹⁾ studied the oxalato complexes of trivalent iron by the method of formation in which the p_H of beginning of precipitate of feric hydroxide in presence of oxalic acid is a function of the concentration of the ferric ion in the free state in solution. This author could detect all the three equilibria ($k_1 = 10^{-9.4}$, $k_2 = 10^{-6.8}$, $k_3 = 10^{-4}$). This shows that once formed bis-oxalato complexes are very stable. For all these reasons, only the k_3 values are considered for comparison. When compared the metals stand as $\text{Al} < \text{Ga} < \text{Fe} < \text{Cr}$ ($k_3^{\text{Al}} = 2.08 \cdot 10^{-4}$, $k_3^{\text{Ga}} = 2.85 \cdot 10^{-5}$, $k_3^{\text{Fe}} = 1.7 \cdot 10^{-5}$, $k_3^{\text{Cr}} = 3.36 \cdot 10^{-6}$). The equilibrium between trivalent chromium ion and oxalato ion has not been studied previously. Values obtained by us for $k_3^{\text{Al, Ga, Fe}}$ differ to certain extents from the values obtained by the previous workers. The methods adapted by the previous workers

¹¹⁾ J. LAMBLING, Bull. Soc. chim. France, Mém. [5] **16**, 495 (1949).

are not free from defects since in those cases the formation as intermediate products hydroxo-complexes cannot be ruled out e. g. $\left[\begin{array}{c} \text{OH} \\ \text{M H}_2\text{O} \\ \text{O}_{x_2} \end{array} \right]''$

and also it is very difficult to detect the exact point at which transparent, gelatinious precipitate of $\text{Al}(\text{OH})_3$ begins to appear; whereas in the present the intermediate products would be aquo-complexes like $\left[\begin{array}{c} (\text{H}_2\text{O})_2 \\ \text{M} \\ \text{O}_{x_2} \end{array} \right]'$ and this will not disturb the system since reaction is carried out in aqueous medium.

Calcutta, University College of Science, Inorganic Chemistry Laboratory.

Bei der Redaktion eingegangen am 28. Mai 1957.